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From conception to fruition: Co-designing a digital exhibit incorporating embodied cognition to encourage young children's computational thinking in a Science Discovery Centre

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Abstract
Theories linked to embodied cognition emphasise the importance of kinaesthetic learning in shaping higher-order cognitive processing. By spreading the cognitive load to other senses while still contributing to central schema, embodied learning can build long-term memory and engagement, especially in young children, leading to improved performance. This approach is particularly suited to computational thinking (CT) which is arguably the critical digital literacy skill of the twenty-first century. This article reports on the co-design of a digital exhibit in collaboration with a Science Discovery Centre (SDC) intended to encourage CT in young children, underpinned by an embodied cognition approach incorporating whole-body actions. Each stage of the exhibit co-design process was informed by research into embodied cognition as well as competencies associated with CT, namely deconstruction, pattern identification, abstraction, and algorithmic thinking, over a 12-month period from conception to fruition. The article describes how theories surrounding embodied cognition and CT were enacted in the realisation process. It concludes by considering the value of co-designing with industry partners such as the SDC, as well as the value of learning designs incorporating kinaesthetic learning especially in relation to abstract concepts such as CT, as embodied knowledge can lead to the construction of enriched mental representations whereby new information is not just seen and heard but is connected to information from the physical environment.

Keywords Embodied cognition · Computational thinking · Decomposition · Pattern identification · Abstraction · Algorithmic thinking · Educational design

Extended author information available on the last page of the article

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Introduction

There is growing evidence that sensorimotor engagement and physical experiences are inexorably linked to children's cognitive development (Mavilidi et al., 2019; Sung et al., 2017). Education theorists have long postulated that interactive experiences help develop children's cognitive functions (Piaget, 1968; Vygotsky, 1967) through processes which combine mental and physical tasks, later expressed in theories such as embodied cognition (Barsalou et al., 2003; Garbarini & Adenzeto, 2004; Shapiro & Stolz, 2019). In addition, computational thinking (CT) has been described as a key twenty-first century thinking skill, necessary to navigate an information society (Ching et al., 2018; Lee et al., 2023). CT draws upon systematic thinking skills, such as problem-solving, from computer science (Wing, 2006). However, while much research into CT has been conducted, relatively few studies have involved younger children (Hu et al., 2023), especially in relation to the learning potential afforded by an embodied cognition approach in the science, technology, engineering and mathematics (STEM) space.

This article reports on the co-design of an interactive digital exhibit between education researchers and a STEM industry partner (Science Discovery Centre) designed to encourage CT in young children via kinaesthetic learning. The study was framed by a desire to embed theorised embodied cognition and CT principles in the exhibit (Abrahamson & Lindgren, 2014; Lee et al., 2023; Nathan & Swart, 2021). The article briefly outlines the co-design process but intentionally focuses more on how the partnership translated theory into practice. It concludes by considering the value of university-industry collaborations, especially in relation to STEM education, the value of theory in informing digital exhibit design, and how theory and collaborations can inform and contribute to enhanced STEM educational outcomes.

Background—embodied cognition

Both Piaget (1968) and Vygotsky (1967) emphasised sensorimotor engagement and physical experiences as an indispensable part of cognitive development, with recurring interactive experiences enabling children to become active respondents in control of their cognitive functions (Lock et al., 1989). Embodied cognition is based on the term of 'affordances' which examines movement-related interactions that stem from the environment, between objects and their related sensorimotor abilities (Garbarini & Adenzato, 2004; Gibson, 2014). Embodied cognition theories state that action and perception are inextricably bound (Barsalou et al., 2003). According to Wilson (2002), cognition is situated, and the environment is integral to manipulating / holding information to be entered into the cognitive system. Cognition then guides action by processing task-relevant inputs and outputs from the environment, based on mechanisms associated with sensory processing and motor control (Wilson, 2002). Glenberg (1997) has argued that

perceptuomotor experiences play a crucial role in the 'encoding of patterns of possible physical interaction with a three-dimensional world' (p. 1), while sensorimotor systems are highly interconnected with cognition, and are stimulated by external signals through body organs and senses such as eyes, ears, and motion (Pulvermüller, 2013). Sensorimotor resources effectively run simulations of the physical world via representing external information or inference drawing (Wilson, 2002).

In understanding embodied cognition as a powerful tool for learning, it is important to examine human cognitive architecture (Mavilidi et al., 2019, 2022). Conscious processing relies on working memory which is limited in duration and capacity in facilitating the learning of novel information (Sweller, 2003). Novel information presented in different modalities (e.g., tactile, auditory, motor, visual), engages relevant sensorimotor functions and can be simultaneously processed by different sub-systems (Risko & Gilbert, 2016), supporting cognitive activities such as memory, emotion, and language (Barsalou et al., 2003). This form of information processing imposes lower cognitive demands and enriched mental schema whereby different sources are collated and organised within the same structure (Goldin-Meadow et al., 2001; Paas & Sweller, 2012). Further, Paas and Sweller (2012) highlight how movements, evolved to acquire information effortlessly, can assist in learning of explicit instruction in areas such as STEM. Therefore, the external environment and bodily interactions mediate the way in which information is stored in the long-term memory as enriched multimodal mental schema, enhancing higher order processes such as problem-solving (Shapiro & Stoltz, 2019). According to Adolph and Hoch (2019), cognitive development is driven by motor development, which relies on body interactions and sensorimotor experiences shaped by the external environment.

Embodied cognition in early childhood education

Mavilidi et al. (2022) have categorised embodied cognition research into three forms: gestures, whole-body movement and physical activity. Gestures involve hand movement. For example, an early study by Valenzeno et al. (2003) reported that preschool children learned concepts of symmetry after observing videos with teachers pointing and tracing while explaining, while Broaders et al. (2007) reported that gesturing while explaining novel mathematics concepts helped young children develop problem-solving strategies. Both Cook et al. (2008) and Ouwehand et al. (2015) report the value of pointing as a long-term memory and recall aid for young children.

Research involving whole-body movements, defined as fine-motor movements located in hands or legs, can extend inner cognitive processes and make abstract thoughts more concrete and easier to process and remember during learning (Mavilidi et al., 2022). For example, Fisher & Brugger (2011) reported that young children performed better on a number line estimation task when they moved across a spatial number line on a dance mat compared to a stable computer while Glenberg (2017) reported actual and imagined manipulation of toys related to a story

enhanced reading comprehension. Similarly, a series of studies examining the effects of task-relevant, gross-motor physical activities during foreign language, geography, science, and mathematics learning reported improved memory performance and high levels of enjoyment, compared to traditional sedentary control groups (Mavilidi et al., 2019). However, the effectiveness of movements for learning relies on the characteristics of the learning task, the learner and the type of movement (Mavilidi et al., 2018b). For example, coping with highly complex physical actions and problem-solving together can potentially impede declarative memory (Mavilidi et al., 2019). Regardless, researchers have argued for unified approaches comprising integrated, embodied, and relevant tasks for pronounced learning benefits (Skulmowski & Rey, 2018).

Computational thinking

Computational thinking (CT) was first alluded to by Papert (1980) who attempted to democratise computer knowledge to a wider body of required twenty-first century knowledge (Angeli & Giannakos, 2020). While there is no widely agreed definition of CT and its operational context (i.e., whether it is stand alone or infused into other subjects; Sung et al., 2017), most studies cite Wing (2006) that 'computational thinking involves solving problems, designing systems, and understanding human behaviour, by drawing on the concepts fundamental to computer science' (p. 33). Further, there is general agreement that CT involves thinking-doing, not just doing (Sung et al., 2017), and should be considered a universal competence for every child (Voogt et al., 2015; Wing, 2006).

The literature differentiates between CT concepts and CT practices. For Hu et al., (2023), CT concepts include sequences, loops, and events while practices refer to problem solving methods such as decomposition, pattern identification, abstraction and algorithmic thinking, although labels applied to processes can vary. Dietz et al. (2019) define decomposition as breaking problems into more manageable pieces while for Valenzuela (2020), decomposition involves understanding constraints, generating solutions, and evaluating strengths and weakness often undertaken before action as a form of mental planning. However, while Pugnali et al. (2017) acknowledge the value in breaking problems down, they note fundamental differences between recognising problems and identifying the steps for solving them. For Lee et al. (2023), pattern identification involves sorting similarities and differences among and within problems; solutions become easier when problems are recognised. Common patterns can then be applied to new problems, making problem solving more systematic and efficient. Lee et al. (2023) describe abstraction as the filtering out of unnecessary detail, ignoring potential distractions, and prioritising the most relevant information while algorithmic thinking involves step by step planning and the generation of rules to follow in the correct order with clear start, end and in-between steps. This is similarly described by Shute et al. (2017) as systematic testing.

In order of thinking processes, Lee et al. (2023) claim that problems are first decomposed, i.e., broken down into manageable smaller problems, from which

patterns of previously solved problems can be identified and applied and unnecessary information discarded. Algorithms then become a map or set of rules to be implemented; sets of algorithms when executed can solve larger problems. While Dietz et al. (2019) state that good CT can help achieve goals effectively and efficiently, Wang et al. (2020) caution for the need to consider developmental appropriateness in relation to task complexity and the nature of the user interface, especially in digital settings.

Computational thinking in young children

Computational thinking now comprises a major part of school curricula (Angeli & Giannakos, 2020), but less attention has been paid to children aged 3–8 (Hu et al., 2023). In their systematic review, Su and Yang (2023) reported only 26 CT research articles involving young children between 2011 and 2022. Aside from the paucity of research, they also note a lack of systematic knowledge as to how to integrate CT into early childhood settings; there is consensus as to what it comprises but not how to apply it. Saxena et al. (2020) claim that children aged 4–6 can demonstrate competencies such as decomposition, pattern identification and abstraction. However, these skills need to be encouraged through the design of age-appropriate digital technologies which accommodate different developmental competencies (Su & Yang, 2023). In this, Wang et al. (2020) note the unevenness of early childhood CT research. Much has been conducted around stand-alone robotics and programming involving debugging, described by Lavigne and Cueller (2019) as identifying problems, brainstorming, applying fixes and checking results. This research largely focuses upon physical outcomes rather than the dynamic learning process (Hu et al., 2023). Finally, robotics and programming research does not necessarily consider interpersonal and intrapersonal attitudes and mind sets such as perseverance (Lavinge & Cueller, 2019) and persistence (Gomes et al., 2018), and presents limitations in offering children the opportunities to practise and develop CT processing in as many ways as possible (Lavigne et al., 2020).

CT & embodied cognition

It is well known that young children learn best through concrete hands-on play (Bers, 2018). While Johnston-Glenberg et al. (2014) proposed that embodied learning supports the learning of abstract concepts and aids retrieval and retention Sung et al. (2017) argued that the design of the learning environment in promoting explicit thinking is key in early childhood education, especially when children engage in full body movement. While acknowledging recent technological advances, Sung et al. (2017) question how to best support children's STEM learning with these technologies.

In their study, Sung et al. (2017) divided 66 preschoolers into four groups involving fully embodied, low embodied and static mathematics activities. They reported that the fully embodied group scored higher in post-numeracy tests suggesting that mental models were strengthened and became more flexible and adaptive as the level

of control over the stimulation and the movement involved expanded. In relation to CT, all competencies improved regardless of embodiment suggesting that just the act of engaging children with CT improved understanding. Hu et al. (2023) engaged children in a range of unplugged embodied CT activities involving surrogate robots whereby children took turns to 'program' as both robot and programmer. They drew upon kinaesthetic CT pedagogies including the physical enactment of pre-defined scenarios with specific cues where students act as robots (Fadjo, 2012; Saxena et al., 2020), and the use of external surrogates whereby learners control and manipulate them as virtual characters or physical objects (Kopcha et al., 2021). Importantly, Hu et al. (2023) reported these embodied activities to have high motivational appeal, and cautioned that past research has focused largely on cognitive outcomes and not subjective feelings. Similarly, preschool children displayed higher numeracy skills when performing activities using foam number blocks placed on a straight line compared to children who sat down and repeated the numbers (Mavilidi et al., 2018a, 2018b). Ruiter et al. (2015) reported that children engaged in whole body activities performed better at math tasks than at a computer. While the encouragement of CT processes undoubtably relates to the design of the activities themselves, the nature of representation and real-time feedback, Kopcha et al. (2021) claim that it is no longer a question of whether learning is embodied but how CT is embodied and how it can be studied through children's interactions with technology. In guiding children's CT process development, they call for immersive educational activities in which mind, body and environment are inextricably linked with and through technology. Their call ultimately informed the current study.

The research partnership

The industry partner for this project was a Science Discovery Centre (SDC) aligned to a wider group of organisations known as GLAMs (Galleries, Libraries, Archives and Museums). GLAMs are physical spaces that contain tangible items such as films, books and exhibits (Riley-Huff et al., 2016), and offer diverse learning opportunities, especially for young children (Duan et al., 2021); the child controls what they engage with, the pace of learning, and no assessment of learning is required (Thuneberg & Salmi, 2018). According to Carr (2000), '[GLAMs] are incendiary cultural institutions [that] feed flames that illuminate the human capacity to imagine the possible' (p. 117), while Griffiths and King (2008) describe them as 'trusted sources of information' that 'serve important complementary roles in supporting a wide variety of information needs' (p. 24). The stated mission of the SDC in this study is to 'increase awareness, interest, capacity and participation in STEM' by encouraging children to ask the 'Why' (Scitech, n.d.).

The value of industry-university partnerships

The importance of industry—university partnerships is acknowledged through the UNESCO First International framework on Open Science which calls for

transparent and accessible international cooperation (UNESCO, 2023) and is an Australian Government priority which calls for STEM collaborations to grow Australia's capabilities (Department of Industry, Science and Resources, n.d.) as part of a wider call for knowledge transfer and cooperative research (Santoro, 2000). Ehrismann and Patel (2015) note interest from industry because they do not always have the research capacity while for universities, collaborations offer the potential to test out ideas, and provide funding and publication opportunities. They note that successful collaborations are based upon mutual trust, shared mission and risk mitigation, and advocate for the perceived value of the collaboration guiding negotiations rather than financial metrics. Similarly, Doley and Kirk (2007) note that industries benefit by outsourcing verification while universities generate new knowledge based on their discovery work.

In terms of collaboration rationales for GLAMs, Yarrow et al. (2008) claim that motivation is driven by the need to maintain community relevance, strengthen public standing, improve services and programs and meet the needs of larger, more diverse cross-sections of learners. They describe three types of partnerships: (1) a focus on collaborative programming, (2) the creation of digital resources, and (3) colocation based upon shared facilities, with each driven by a desire to support life-long learning and/or community development, optimise services provision, enable universal access, broaden customer bases, and/or preserve heritage materials. However, they acknowledge the lack of research in this area, particularly into collaborative rationales for SDCs (Yarrow et al. 2008).

Designing the digital exhibit

Educational design

Educational design is a relatively new area defined as the process of designing evidence-driven, technology-based learning environments hypothesised to influence student thinking, and involves taking science-based theories and creating models for applied usage (Nathan & Swart, 2021). Intentionally bridging a gap between theory and practice, Nathan and Swart (2021) refer to it as 'learning engineering' (p. 1931). They differentiate between complex systems and complicated systems whereby complex systems incorporate subsystems making design more systematic, while complicated systems arise more organically. Complex systems utilise conceptual frameworks (such as embodied cognition) so that 'theoretical considerations of learning…in complex neural, cognitive… and cultural systems will yield critically important insights of central relevance' (Jacobson et al., 2016, p. 217). However, as theories underpinning educational design have not always been empirically established across all settings, Nathan and Swart (2021) advocate a process of constant oscillation between theory and practice.

Embodied cognition design principles

In considering embodied cognition as the design framework for this project, Abrahamson and Lindgren (2014) state that '…all STEM content, [is] grounded not in its sign systems and inscriptional forms…but in the situated, spatial-dynamical, and semantic phenomenology of the person who is engaging in the activity…' (p. 358). They note a difference between 'immediate doing and mediated thinking', whereby absorbed experience is better understood through structured reflection, marking the difference between effortless intuition and deliberate reasoning. Accordingly, they pose the first of three overarching design challenges: (1) how can learning environments be designed which challenge children to rethink as they undertake the activity (even thinking about an abstract concept can stimulate 'thinking action' whereby the physical outcome is imagined without actual doing), (2) how to select and facilitate physical interactions that give rise to thinking aligned with desired educational outcomes, and (3) whether it is possible to reapply what has been learned when technology is absent. The three challenges relate to activities, materials and facilitation, and enabling conditions associated with each are set out in Table 1.

In connection with activities, Sung et al., (2017) suggest initial consideration of the degree of embodiment, i.e., gesture, whole body or physical, as well as the nature of embodiment, i.e., direct or surrogate. In the latter, the learner controls the movements of an external surrogate such as a virtual character thereby replicating the thought processes of a programmer. Operating the surrogate requires children to display explicit knowledge in the form of commands, imagine the expected performance of the surrogate and transform the movement required into commands. Additionally, activities need to cater for a range of ages and CT potential (Su & Yang, 2023; Wang et al., 2020) and not just reply on software but promote elements of self-discovery (Sung et al., 2017).

For materials, Mavilidi et al. (2018a) and Pesce (2012) note the qualitative importance of task novelty, effort, and mental engagement, while Pesce et al. (2019) claim task diversity, novelty, effort and mental engagement to be underpinning mechanisms for transforming physical activity into meaningful learning experiences, through intrinsic motivation. In practical terms, Pugnali et al. (2017) notes the need to consider proportionality, i.e., physical size of participants, while the use of metaphors offers a purpose for engagement (Manches et al., 2020). Finally, Tomporowski and Qazi (2020) caution consideration of task complexity and mental demands, i.e., coping with both physical actions and problem-solving together, as under certain conditions, this may impede declarative memory.

In relation to facilitation, Nathan and Swart (2021) acknowledge learning to be complex, non-linear and reliant on feedback and feedforward for self-regulation, and therefore state the need for real time feedback for ongoing engagement. Sung et al. (2017) state that higher levels of feedback support children's perceptual experience in concrete simulations to understand abstract concepts. However, Pentimonti and Justice (2010) caution as to the amount of scaffolding required, with less directed guidance promoting better higher-order thinking. As cognitive systems are essentially predictive, Nathan and Swart (2021) and Kopcha et al. (2021) advocate the inclusion of alternative pathways that encourage trial and error with positive,

Table 1 Enabling conditions associated with embodied cognition design (Abrahamson & Lindgren, 2014)

Activities	Materials	Facilitation
Draw upon children's pre-existing capacity in a real or 3D space	Fully use available media	Patterns of movement may not be instinctive—need guidance as to how to move their bodies and spatial relations (functional metaphors)
Use children's senses and coordination to judge stimuli to perform new actions	Set a purpose for engaging with the materials similar to the non-digital world	Educators and experts work together to overcome 'hidden' aspects of the activity
Focus not on the STEM problem but the figurative representation	Set an action-feedback loop	Opportunity for demonstration, coaching
Start simple and move to more complex objectives over time	Experience first, and the manipulation of the virtual object (no matter how small the action)	Opportunities for children to articulate the strategies they used
	Gradually introduce objectives that require new conceptual configurations/tools of reference to develop new perceptuomotor schemas	

physical feedback. Finally, Kopcha et al. (2021) report the value of grids as a design reference for encoding bodily understanding to help with symbolic manipulation by better connecting bodily experience with proportionality. While acknowledging the richness of design principles set out in the literature, few have tried to implement them systematically, with the exception of Hu et al. (2023), Sung et al. (2017) and selected others.

Computational thinking design principles

In considering design principles associated with children's CT processing, these have been thoroughly summarised by Lee et al. (2023) and are presented in Table 2.

Having considered the literature informing design principles associated with embodied cognition and CT, the article now briefly outlines the co-design process, and sets out how these the principles were embedded in the built digital exhibit.

The co-design process

For this project, co-design involved a five-stage process advocated by Bahr and Bolton (2002) as: (1) pre-planning, (2) planning, (3) implementation, (4) evaluation and (5) sharing the experience, of which this article focuses on the physical outcome of the first three. However, in terms of process, it was the industry partner who initiated the collaboration, and creation involved the cross pollination of ideas between the university researchers who led discussion in terms of learning theory, and the industry partner in terms of technical expertise with the express aim of creating a digital resource. This occurred via monthly meetings between all stakeholders over a 12-month period from conception to completion of the exhibit prototype. Numerous ideas were shared and considered across all stages (Nathan & Swart, 2021) guided by the call by Kopcha et al. (2021), as to how best utilise the phenomenon of embodiment in an immersive technology-supported CT experience.

The digital exhibit (product)

The built digital exhibit ultimately comprised a complex system (Nathan & Swart, 2021) involving both direct and surrogate full body movement (Sung et al., 2017). It employed a tangible coding floor mat which incorporated a series of colour coded directional arrows activated by stepping on them (including an eraser function). Each arrow controlled the directional movement of a mouse on a large digital screen facing the dance mat. The goal was for children to help the mouse navigate on-screen mazes at ever increasing levels of complexity to reach a piece of cheese. Both the large digital screen and the floor mat were created with the proportionality of 3–8-year-olds in mind (Pugnali et al., 2017) and the use of a metaphorical mouse (Manches et al., 2020) to generate real-world relatability, novelty and motivation. The exhibit comprised 12 game levels to accommodate a range of ages and

Table 2 Thinking dispositions associated with CT (Lee et al., 2023)

Decomposition	Pattern identification	Abstraction	Algorithmic thinking
Activity has components or pieces	Activity has similarities within or to other activities	Activity has information that is irrelevant or not useful	Activity has rules or a step by-step process
Increase/decrease the number of components or elements in an activity	Increase/decrease the length of the pattern or sequence	Increase/decrease the number of relevant pieces of information	Increase/decrease the number of steps or rules needed
Represent components or elements in a written list or in visual form	Increase/decrease the similarities or differences between activities	Increase/decrease the number of irrelevant or not useful pieces of information	Change the possible order of steps for a successful outcome
What are the steps to complete this problem?'	'How is this similar to something else we've done?'	'What are the important details?'	'What are the rules we must follow?'

abilities (Wang et al., 2020), and each game offered participants three 'lives', after which it defaulted back to level 1. Children could work through each level at their own pace (Nathan & Swart, 2021), and higher game levels incorporated alternative pathways (Kopcha et al., 2021). Each game level was proceeded by an onboarding video which visually demonstrated how the respective level worked, without verbal instructions to reduce overreliance on scaffolding and promote higher order thinking (Pentimonti & Justice, 2010), and real time as well as post-game feedback (Nathan & Swart, 2021) to support self-discovery and engagement (Sung et al., 2017). In terms of CT, each level required initial decomposition and pattern identification, while successive levels incorporated greater abstraction while building algorithmic thinking through the application of underlying pathway 'rules' (Lee et al., 2023). To not overload declarative memory, the exhibit fundamentally required children to think first and then act (Abrahamson & Lindgren, 2014; Tomporowski & Qazi, 2020). Figure 1 illustrates the tangible coding floor mat and screen in action.

Embodied cognition

In terms of embodied cognition, Table 3 maps the digital exhibit against both embodied cognition design challenges presented by Abrahamson and Lindgren (2014).

These challenges functioned as a checklist during both the design and build as the researchers and the SDC technical team consciously addressed each affordance.

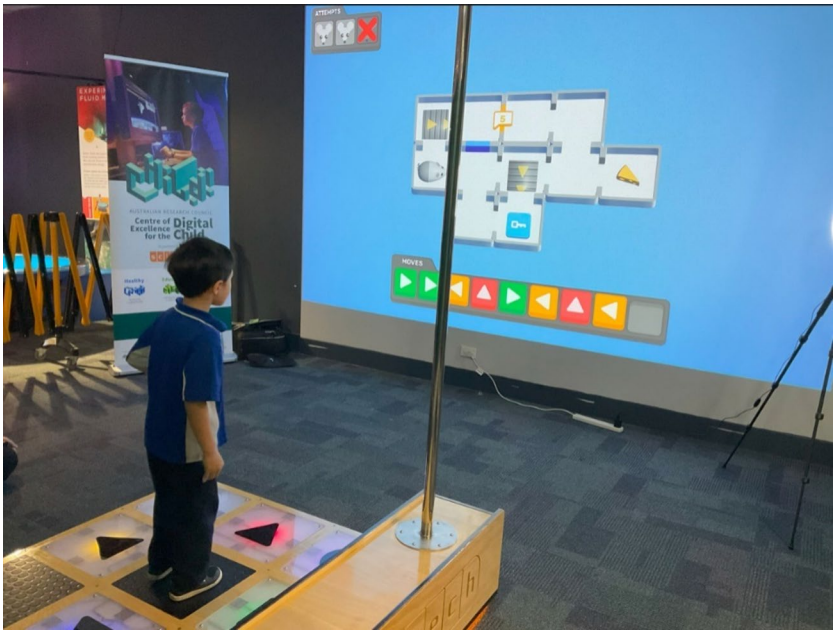


Fig. 1 Digital exhibit product in action

Table 3 Digital exhibit embodied cognition enabling conditions

Challenges	Enabling conditions	Digital exhibit
Activities	Use children's pre-existing capacities	Exhibit employed both visual and motor capabilities
	Use senses and coordination to perform new actions	Exhibit drew upon children's ability to apply visual and motor coordination (stepping)
	Focus not on the STEM problem but figurative representation	Exhibit employed the visual metaphor of a mouse trying to get to a piece of cheese
	Start simple and move to more complex over time	Exhibit involved 12 levels of increasing complexity
Materials	Fully use available media	Exhibit involved large digital screen for representation of a maze and a dance mat for maze activation
	Provide a purpose for engaging similar to the non-digital world	Purpose was provided via helping a mouse get a piece of cheese to eat
	Need for action-feedback loop	Each action was followed by a real time feedback loop indicating success or failure
	No matter how small, experience firsthand the manipulation of the virtual object	Digital mouse responded exactly to manipulated motor-initiated commands
	Gradually introduce objectives that require new conceptual configurations / tools of reference to develop new perceptuomotor schemas	Exhibit involved 12 levels of increasing complexity, each of which introduced new conceptual configurations such as gates, elevators and slides
Facilitation	Patterns of movement may not be instinctive—need guidance as to how to move their bodies and spatial relations	Each exhibit level was preceded by an onboarding screen with initial screen outlining spatial relations between digital maze and dance mat
	Educators and experts work together to overcome 'hidden' aspects of the activity	The 12-month design/build allowed time to test & solve educational and technical issues
	Opportunity for demonstration, coaching	Onboarding screens at each level demonstrated level procedures, and the physical exhibit design offered opportunities for sideline coaching
	Opportunities for children to articulate the strategies they used	Exhibit included real time representation of moves as well as overall strategies

Computational thinking

In terms of CT, Table 4 maps the digital exhibit against the thinking dispositions described by Lee et al. (2023).

As with embodied learning enabling conditions, CT dispositions were employed as an overarching outcomes framework which guided the exhibit design and build at all stages. The finished exhibit was initially tested in June 2022 in closed sessions with children aged 4–8, and then placed on the SDC exhibit floor in January 2023 for general public use.

Discussion

At the outset, this article has set out to illustrate the processes that informed this partnership, namely the creation of a digital exhibit using theoretical principles derived from embodied cognition to encourage CT in young children. As such, it has attempted to highlight the value to both parties of this form of collaboration. As noted by Ehrismann and Patel (2015), industry partners often have the technical knowhow but lack the theoretical knowledge, in this case, in an education context. The university researchers benefited from the ability to test out theories in a faster time frame and real world setting while the industry partner benefited from the education theory underpinning exhibit creation, driven by its need to maintain relevance in meeting the learning needs of its audience (Yarrow et al., 2008) while addressing its mission of 'increasing awareness, interest, capacity and participation in STEM by encouraging children to ask the why' (Scitech, n.d.). In reflection, the collaboration has been deemed successful because it was driven by the perceived value of exhibit creation, rather than financial benefit, to achieve both organisations shared mission of encouraging children's interest in STEM. Further, the collaboration has benefited from a co-design process based around mutual trust and clear development frameworks established at the beginning of the project (Ehrismann & Patel, 2015), supported by ongoing regular dialogue as recommended by Bahr and Bolton (2002). As such, the authors advocate this approach as a potential collaboration framework for future GLAM-university partnerships (Yarrow et al., 2008).

In terms of the co-design process from conception to fruition, the collaboration benefitted from an in-depth exploration of the embodied cognition, CT and education design literature at the preplanning and planning stages. Accordingly, theory was able to guide exhibit development through a clear understanding of the desired outcomes (CT), the processes required to enable the outcomes (embodied cognition) and the processes surrounding exhibit build (educational design). As such, the creation of design principles, presented as tables in this article, proved invaluable in planning and reflection as points of reference throughout the build, and highlighted the role of theory in the construction of complex systems (Nathan & Swart, 2021). While the university partner provided the theoretical frameworks that guided construction, it was the technical expertise of the SDC builders who were able to shape them into something concrete, and this highlights inherent limitations for education researchers with theory per se—theories are, in fact, just theories which

Table 4 Digital exhibit CT thinking dispositions

CT elements	Thinking dispositions	Digital exhibit
Decomposition	Activity has components or pieces	Activity involved observing, mental planning, facilitation and feedback
	Increase/decrease number of components or elements	Exhibit involved 12 levels of increasing complexity
	Represent components/elements in list or visual form	Moves were presented in visual form on both the dance mat and digital screen
	What are the steps to complete this problem?	Children required to break down (mentally pre-plan) required steps at each level
Pattern Identification	Activity has similarities within or to other activities	Exhibit involved concept of a maze common in education pedagogy
	Increase/decrease pattern length or sequence	Sequences increased in length at higher game levels
	Increase/decrease similarities or differences	Successive levels used similar principles while introducing differences to be solved
	'How is this similar to something else we've done?'	Each game level incorporated similar principles from preceding game level
Abstraction	Activity has information that is irrelevant or not useful	Activity did not employ irrelevant information, but possible traps at higher levels
	Increase/decrease relevant pieces of information	Each game level incorporated similar principles from previous game levels
	Increase/decrease irrelevant pieces of information	Each game level introduced new challenges built upon existing information
	'What are the important details?'	Each game level required more focused thing to ignore possible distractions
Algorithmic thinking	Activity has rules or a step by-step process	Each game level was preceded by an onboarding screen indicating processes
	Increase/decrease number of steps or rules needed	Each game level introduced new rules built upon existing rules from preceding levels
	Change the possible order of steps for successful outcome	Higher game levels incorporated multiple pathways to achieve outcomes
	'What are the rules we must follow?'	Each game level employed similar rules which became more complex at successive levels

can be difficult to implement given the fluidity of underlying constructs and lack of empirically defined boundaries. As noted by Nathan and Swart (2021), a co-design process is not linear and requires constant reflection and re-evaluation between partners over the build time; for example, this project required a complete rethink of the earliest exhibit designs when mapped against the theoretical frameworks. This again highlights the importance of dialogue and respect for each partner's perspectives in the complex translation of theory to practice, and the need for clear guiding frameworks. For this project, the three design questions posed by Abrahamson and Lindgren (2014) proved particularly valuable and functioned as constant points of reference surrounding the degree of embodied cognition enactment, proportionality, novelty and motivational appeal of the exhibit to children, and the developmental functionality of the exhibit in achieving CT outcomes. The ability to compartmentalise and reflect on each within the dynamic process of creation undoubtedly contributed towards a successful material outcome.

It is not the role of this article to discuss the empirical properties of the project in terms of encouraging CT. This will be reported separately elsewhere. The purpose of this article has been to highlight the co-design processes, from the clearly defined design stages (Bahr & Bolton, 2002) and the principles underpinning the collaboration (Ehrismann & Patel, 2015), to the value of theoretical frameworks in guiding the build (Abrahamson & Lindgren, 2014) and an understanding of the oscillation between theory and practice as a normal part of the education design build process (Nathan & Swart, 2021). Regardless, the authors are confident that any success in terms of CT outcomes associated with the exhibit can be directly attributed to the co-design processes described here. Since being open to the public, both SDC staff and the researchers have observed that children have been highly engaged and appear motivated by the materiality of the exhibit which, by association, must be meaningfully impacting their CT development (Mavilidi et al., 2018a, 2018b; Pesce et al., 2019). Therefore, the authors advocate for the processes set out in this paper in addressing the call for immersive educational activities in which mind, body and environment are inextricably linked with and through technology (Kopcha et al., 2021).

Conclusion

Embodied cognition is increasingly recognised as a powerful learning approach in shaping higher order cognitive thinking. In addition, CT is widely acknowledged as a critical twenty-first century literacy skill. It stands to reason that embodied cognition is well suited to encouraging CT in the digital STEM space, especially in helping make abstract concepts explicit for young children. The issue has not been whether to do it, but how (Kopcha et al., 2021). The creation of the purposefully designed and built digital exhibit described in this article is but one example of translating theory into practice. In line with theoretical design principles, the resulting digital exhibit combined perceptuomotor and sensorimotor experiences by encouraging thinking about physical interactions and responding to whole body stimulation. Information was presented in different modalities namely sight, sound

and motion in ways which were deliberately designed to stimulate memory, emotional connection and thinking-doing around immediate doing and mediated thinking, in a task-appropriate way for young children.

Of equal importance has been the process of creation, in terms of acknowledging the strengths and expertise each partner brings to the co-design process, the co-design pathway and the shared partnership objectives. Having clear frameworks in place, commencing with a detailed examination of the design principles inherent in the literature, a clear understanding of the educational design pathway, trust, and open and transparent communication has resulted in an exhibit which the SDC acknowledges as having accelerated its understanding of digital exhibit design by at least two to five years. Formulating clear partnership expectations at the outset, focusing on open communication, and having a transparent co-design process has been central to this partnership, suggesting that these processes are as important as the aims and objectives of any partnership. Importantly, this partnership goes partway to addressing calls to encourage CT through embodied interactions with technology (Kopcha et al., 2021) and the broader statement by Diamond and Ling (2016) that 'for better learning outcomes, children should be exposed to holistic learning experiences [that] engage them physically, cognitively, psychologically and socially' (p. 43), at least in the GLAM STEM space.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Karen Murcia, Emma Cross, Michelle de Kok and William Peng. The manuscript was lead and co-written by Geoffrey Lowe, with contributions from Myrto Mavilidi and all authors contributed feedback in the process. All authors read and approved the final manuscript.

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Data availability Data will be made available on request that is allowable within the project's university ethics conditions.

Declarations

Conflict of interest The research team are all members of the ARC Centre of Excellence for the Digital Child. William Peng is the General Manager Exhibitions at Scitech, Perth Western Australia's Science Discovery Centre. Scitech is an Industry Partner with the ARC Centre of Excellence for the Digital Child.

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